TUTORIAL 14

Selenium Automation: Login, Add Product to Cart, and Validate

2. Requirements

- Python installed on the computer.
- Google Chrome browser installed.
- Visual Studio Code (VS Code).
- Selenium Python package.

3. Steps to Perform the Practical

1. Open VS Code and create a Python file named tutorial14.py.
2. Open the VS Code terminal and install Selenium by running: pip install selenium
3. Make sure Google Chrome is installed. Recent Selenium versions normally manage the required driver automatically.
4. Copy the complete code from Section 4 into tutorial14.py and save the file.
5. Run the program using the Run button in VS Code or the command: python tutorial14.py
6. The script opens SauceDemo, logs in, adds the backpack to the cart, validates the cart, prints the results, and closes the browser.

4. Complete Python Code

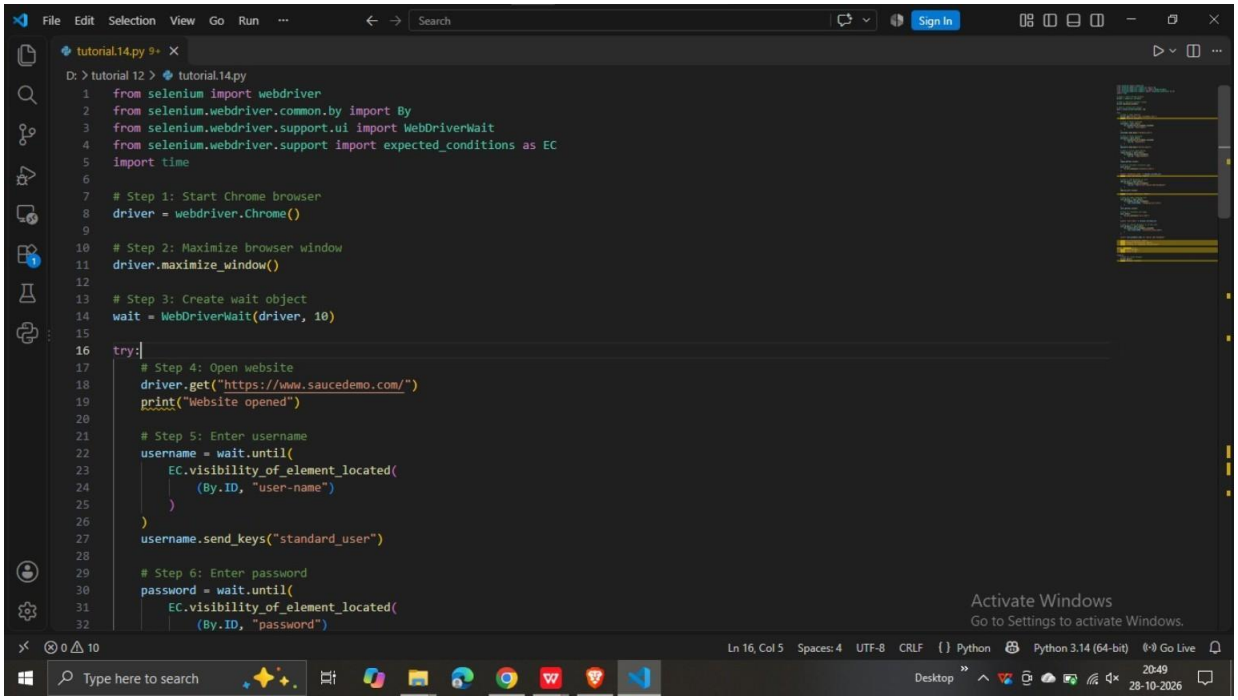
```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
import time
# Step 1: Start Chrome browser
driver = webdriver.Chrome()
# Step 2: Maximize browser window
driver.maximize_window()
# Step 3: Create wait object
wait = WebDriverWait(driver, 10)
try:
    # Step 4: Open website
    driver.get("https://www.saucedemo.com/")
    print("Website opened")
    # Step 5: Enter username
    username = wait.until(
        EC.visibility_of_element_located((By.ID, "user-name"))
    )
    username.send_keys("standard_user")
    # Step 6: Enter password
    password = wait.until(
        EC.visibility_of_element_located((By.ID, "password"))
    )
    password.send_keys("secret_sauce")
    # Step 7: Click Login button
    login_button = wait.until(
        EC.element_to_be_clickable((By.ID, "login-button"))
    )
    login_button.click()
```

```
    # Step 8: Validate inventory page
    wait.until(EC.url_contains("inventory.html"))
    assert "inventory.html" in driver.current_url
    print("Login Validation: PASS")
    # Step 9: Add Backpack to cart
    add_to_cart = wait.until(
        EC.element_to_be_clickable(
            (By.ID, "add-to-cart-sauce-labs-backpack")
        )
    )
    add_to_cart.click()
    print("Product Selection: PASS")
    # Step 10: Open shopping cart
    cart_button = wait.until(
        EC.element_to_be_clickable((By.CLASS_NAME, "shopping_cart_link"))
    )
    cart_button.click()
    # Step 11: Validate cart page
    wait.until(EC.url_contains("cart.html"))
    assert "cart.html" in driver.current_url
    print("Cart Validation: PASS")
    # Step 12: Verify product in cart
    product = wait.until(
        EC.visibility_of_element_located((By.CLASS_NAME, "inventory_item_name"))
    )
    assert product.text == "Sauce Labs Backpack"
    print("Product Verification: PASS")
    print("Tutorial 14 Completed Successfully")
except Exception as e:
    print("Tutorial 14: FAIL")
    print("Error:", e)
finally:
    # Step 13: Close browser
    driver.quit()
    print("Browser closed")
```

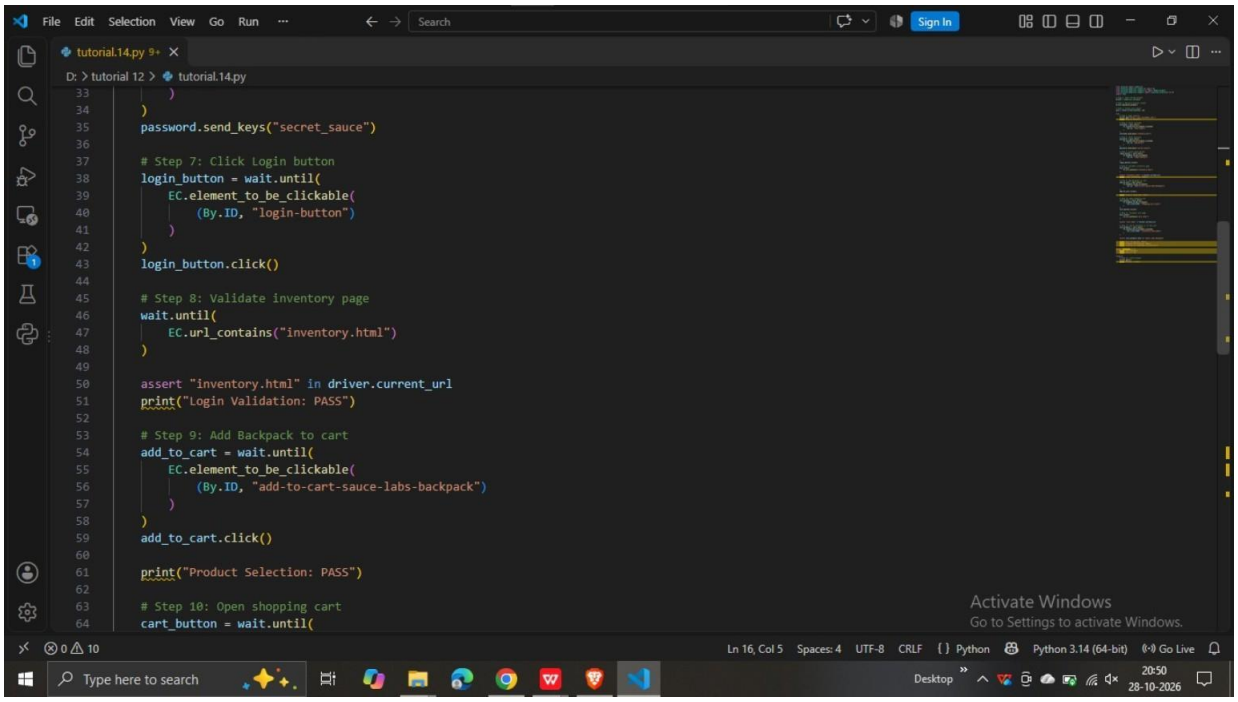
5. Code Explanation

- **webdriver.Chrome() —** Starts a Chrome browser controlled by Selenium.
- **WebDriverWait(driver, 10) —** Waits up to 10 seconds for an element or condition.
- **By.ID —** Locates a webpage element by its HTML id.
- **send_keys() —** Types the username and password into the login fields.
- **click() —** Clicks the Login, Add to cart, or cart button.
- **EC.url_contains() —** Waits until the browser URL contains the expected page name.
- **assert —** Checks that the expected page or product is present; otherwise, an exception is raised.
- **try / except / finally —** Handles errors and ensures that the browser is closed at the end.

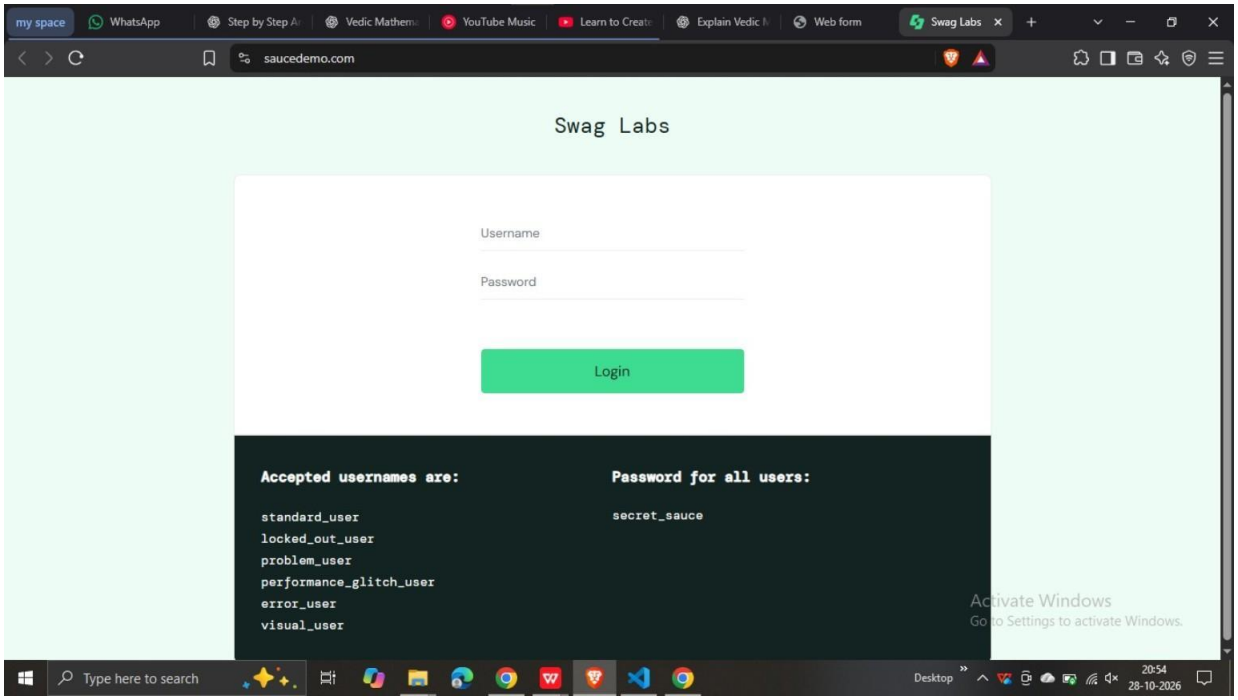
6. Screenshots



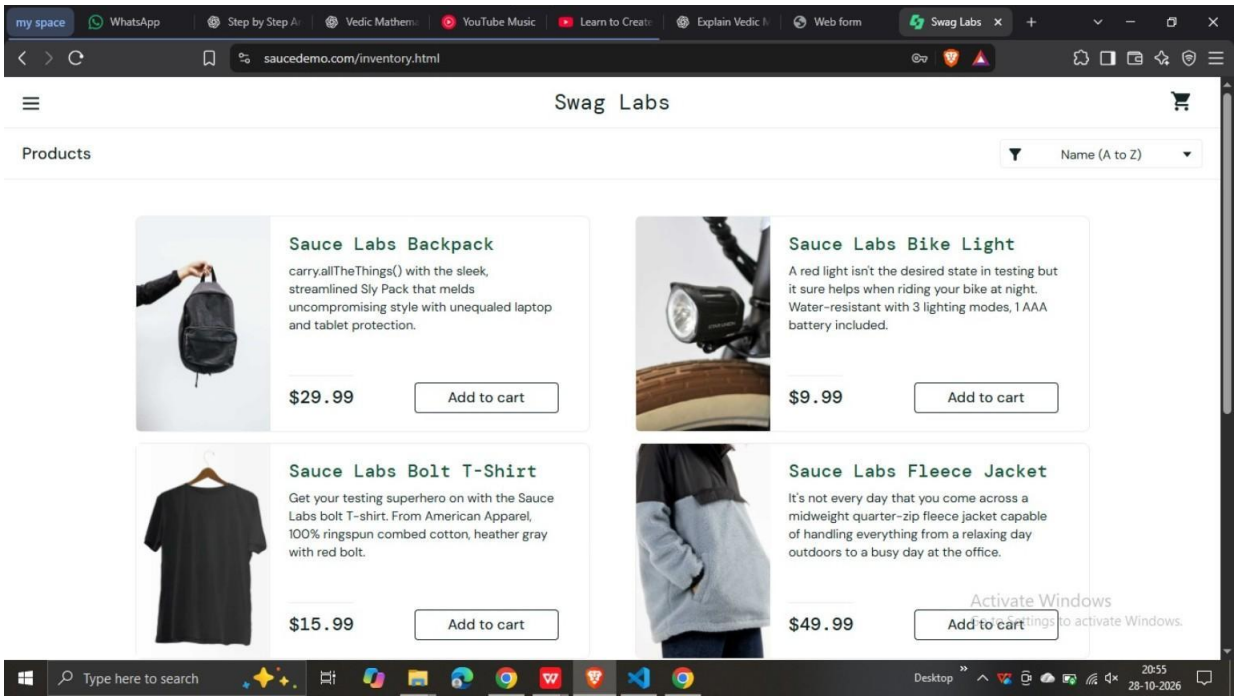
Screenshot 1: Python code — browser setup and login steps.



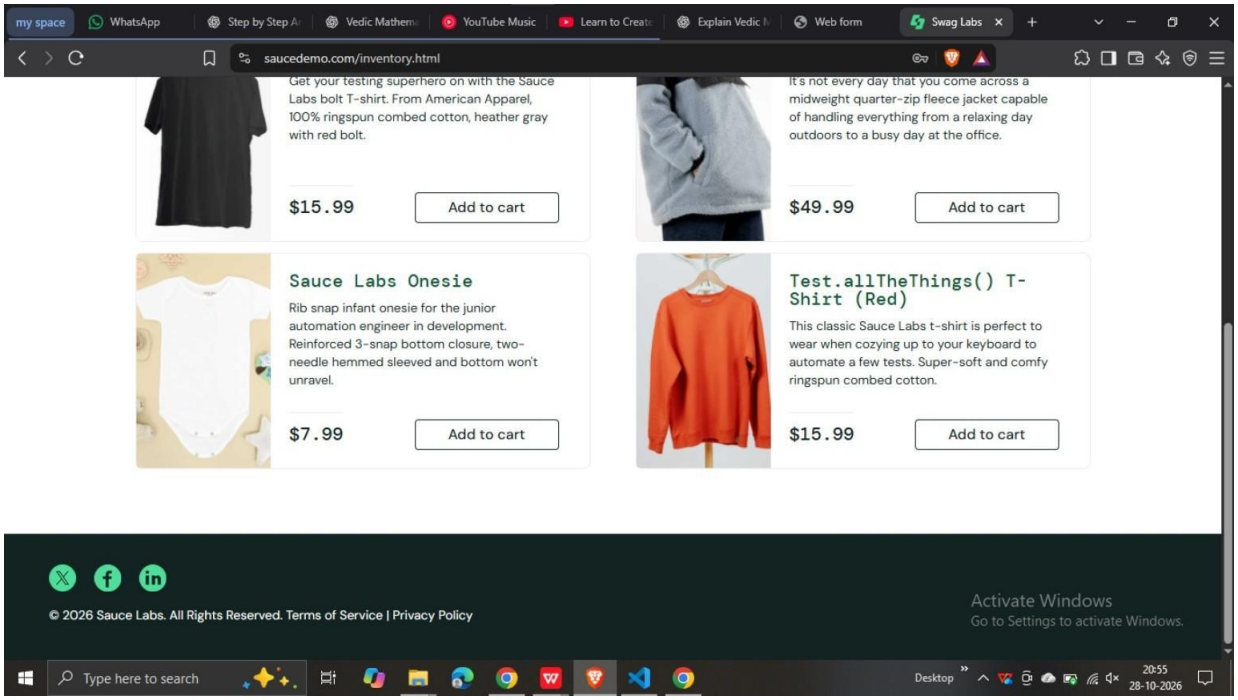
Screenshot 2: Python code — cart validation and product selection section.



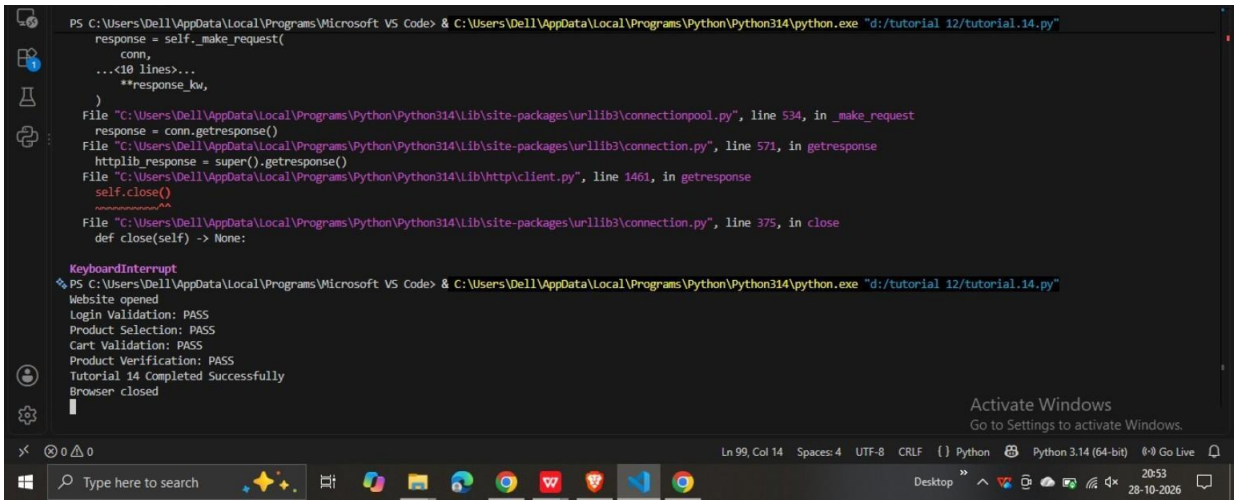
Screenshot 3: SauceDemo login page.



Screenshot 4: SauceDemo inventory page with products.



Screenshot 5: Lower part of the inventory page.



Screenshot 6: VS Code terminal showing the PASS messages and successful completion.

7. Expected Output

```
Website opened
Login Validation: PASS
Product Selection: PASS
Cart Validation: PASS
Product Verification: PASS
Tutorial 14 Completed Successfully
Browser closed
```

8. Result / Conclusion

The SauceDemo shopping workflow was automated using Python and Selenium. The program opened the website, logged in with the test credentials, added the Sauce Labs Backpack to the cart, verified the cart contents, printed PASS messages, and closed the browser.

Note: The code in this document is reconstructed from the visible screenshots. The screenshot does not show every line of the original file, so the cart-validation and final cleanup lines are written to match the successful output shown.